

CS302P - Digital Logic Design (Practical) (Semester Quiz # 03)

Question # 1 of 10 (**Start time: 03:36:09 PM, 22 December 2025**)

_____ gate with a bubble at the output is known as NOR gate.

Select the correct option

☐

NAND

☐

AND

☐

NOT

☒

OR

Question # 2 of 10 (start time: 03:36:32 PM, 22 December 2025)

A full adder circuit takes _____ bit(s) as input and returns _____ bit(s) as output.

Select the correct option

- | | |
|----------------------------------|------|
| ☐ | 3, 1 |
| ☐ | 2, 1 |
| ☐ | 2, 2 |
| ☒ | 3, 2 |

Question # 3 of 10 (**start time: 03:36:51 PM, 22 December 2025**)

_____ is also known as a Universal gate.

Select the correct option

☐	AND
☐	OR
☐	XNOR
☒	NOR

Question # 4 of 10 (**start time: 03:37:15 PM, 22 December 2025**)

There are _____ cells in a 4-variable K-map.

select the correct option



16



24



20



8

Question # 5 of 10 (Start time: 03:37:39 PM, 22 December 2025)

Total number of inputs in a half adder is _____.

Select the correct option



1



3



4



2

Question # 6 of 10 (**Start time: 03:37:58 PM, 22 December 2025**)

A Karnaugh map (K-map) is an abstract form of _____ diagram organized as a matrix of squares.

Select the correct option

☒	Venn Diagram
☐	Cycle Diagram
☐	Triangular Diagram
☐	Block diagram

Click to

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Question # 7 of 10 (**start time: 03:38:21 PM, 22 December 2025**)

Which of the following is not a valid number for grouping of terms in K-Map?

Select the correct option

☐

1

☐

4

☐

2

☒

3

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Question # 8 of 10 (Start time: 03:38:39 PM, 22 December 2025)

Which of the following gates is used for parity generation for a binary data string?

Select the correct option

☐	AND
☐	OR
☐	NOR
☒	XOR

Question # 9 of 10 (Start time: 03:38:59 PM, 22 December 2025)

In 8x1 multiplexer, which one of the following lines must always be connected to ground in order to activate the multiplexer?

Select the correct option



G'



V_{CC}



Y



W

Click to Save Answer & Move

Question # 10 of 10 (Start time: 03:39:18 PM, 22 December 2025)

In a full adder circuit having inputs **A = 1**, **B = 1**, **Cin = 0**, the outputs **S** and **Cout** will be _____.

Select the correct option

- | | |
|----------------------------------|-----------------|
| ☐ | S = 1, Cout = 1 |
| ☐ | S = 0, Cout = 0 |
| ☒ | S = 0, Cout = 1 |
| ☐ | S = 1, Cout = 0 |

Question # 1 of 10 (**Start time: 03:40:53 PM, 22 December 2025**)

In order to build an XOR gate using NAND gates only, _____ gate(s) are required.

Select the correct option

☐	1
☒	4
☐	2
☐	3

Question # 2 of 10 (**start time: 03:41:15 PM, 22 December 2025**)

When implementing a half adder by using gates, which of the following gates is used to give Sum (S) as an output?

Select the correct option

☒	XOR
☐	NOR
☐	OR
☐	AND

Question # 3 of 10 (**Start time: 03:41:34 PM, 22 December 2025**)

In a full adder circuit having inputs **A = 1, B = 1, Cin = 0**, the outputs **S** and **Cout** will be _____.

Select the correct option

☒	S = 0, Cout = 1
☐	S = 1, Cout = 1
☐	S = 0, Cout = 0
☐	S = 1, Cout = 0

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Question # 4 of 10 (**Start time: 03:41:57 PM, 22 December 2025**)

_____ is also known as a Universal gate.

Select the correct option

☐	OR
☐	XNOR
☐	AND
☒	NOR

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Question # 5 of 10 (Start time: 03:42:16 PM, 22 December 2025)

_____ gate with a bubble at the output is known as NAND gate.

Select the correct option

☒	AND
☐	NOR
☐	NOT
☐	OR

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Question # 6 of 10 (**start time: 03:42:34 PM, 22 December 2025**)

How many select lines would be required for an 8x1 multiplexer?

Select the correct option

☒	3
☐	2
☐	1
☐	8

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Question # 7 of 10 (**start time: 03:42:53 PM, 22 December 2025**)

Which of the following expressions represents S output in a full adder?

Select the correct option



$$A \oplus B \oplus C_{in}$$



$$(A \times C_{in}) \oplus B$$



$$(A + B) \times C_{in}$$



$$(A \times B) \oplus C_{in}$$

Question # 8 of 10 (**Start time: 03:43:26 PM, 22 December 2025**)

Which of the following gates is used for parity generation for a binary data string?

Select the correct option

☒	XOR
☐	NOR
☐	AND
☐	OR

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Question # 9 of 10 (**start time: 03:43:45 PM, 22 December 2025**)

There are _____ cells in a 4-variable K-map.

Select the correct option

☒	16
☐	8
☐	20
☐	24

Question # 10 of 10 (Start time: 03:44:03 PM, 22 December 2025)

Total Marks:

_____ gates are economical and easier to fabricate and are the basic gates used in all IC digital logic families.

Select the correct option

 Reload Math Equations☐

Inverse

☐

Boolean

☐

Logic

☒

Universal

Click to Save Answer & Move to Next Question